

Software Engineering Capstone - gRPC APIs

Duration: 10 minutes.

* Indicates required question

1. Email *

2. Full name *

3. Student ID *

4. Suppose a programming language uses 8 bytes to store each character and * 1 point
4 bytes to store an integer. How many bytes would a program need to
allocate to store the values of two object properties — Name =
"SoftwareEngineeringCapstone" and Year = **2024** — using **text**
encoding?

Mark only one oval.

☐ 31

☐ 220

☐ 224

☐ 232

☐ 248

5. Assume that a programming language uses 8 bytes to store a character and * 1 point
4 bytes to store an integer. How many bytes does a program need to allocate
to store two properties of an object — Name =
"SoftwareEngineeringCapstone" and Year = **2024** — using **binary
encoding**?

Mark only one oval.

- ☐ 31
☐ 220
☐ 224
☐ 232
☐ 312

6. Which of the followings are binary file formats? * 1 point

Check all that apply.

- ☐ .jpg
☐ .zip
☐ .pdf
☐ .exe
☐ .dll
☐ .xml
☐ .js

7. Serialization is ____ . * 1 point

Mark only one oval.

- ☐ to convert programming language object (in memory) to JSON string
☐ to convert JSON string to programming language object (in memory)
☐ to convert programming language object (in memory) to a specific format (e.g. XML, JSON or Protocol Buffer)
☐ to convert a specific format (e.g. XML, JSON or Protocol Buffer) to programming language object (in memory)

8. HTTP/2 is ____ . *

1 point

Mark only one oval.

- ☐ text protocol
- ☐ binary protocol

9. HTTP/2 is ____ . *

1 point

Mark only one oval.

- ☐ stateless protocol
- ☐ stateful protocol

10. gRPC API runs on top of ____ . *

1 point

Mark only one oval.

- ☐ Application layer.
- ☐ Transport layer.
- ☐ Network layer.

11. What are the basic steps to implement gRPC APIs *

1 point

Check all that apply.

- ☐ Select a platform.
- ☐ Select a framework.
- ☐ Import HTTP/2 server and support libraries.
- ☐ Create .proto file.
- ☐ Generate server skeleton and client stub from .proto file using protoc compiler.
- ☐ Implement API functions.
- ☐ Implement routing for each URL (map URLs to API functions).
- ☐ Start local API server.
- ☐ Invoke local APIs.
- ☐ Deploy APIs to production server.
- ☐ Invoke production APIs.

12. When should gRPC APIs be used? *

1 point

Check all that apply.

- ☐ Performance is important.
- ☐ Ease of use is important.
- ☐ Error handling is important.

13. gRPC APIs can be invoked directly from a browser. *

1 point

Mark only one oval.

- ☐ True
- ☐ False

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